

Day 11 : Permission Handling

1. Introduction to Permission Handling

Permission Handling is used to get access to device features like Camera, Location, Storage, Microphone, and Contacts. Flutter apps must request permission from the user before using these resources.

2. Why Do We need Permission?

- Protect user privacy
- Secure device data
- Control access to hardware features
- Follow Android and iOS security policies

3. Common Permission Types

Permission	Purpose
Camera	Take photos/videos
Gallery/Photos	Access images
Location	Get user's location
Storage	Read/write files
Microphone	Record audio
Contacts	Access phone contacts

4. Install Package

dependencies:

permission_handler: ^11.3.1

Run :

Flutter pub get

5. Import Package

```
import 'package:permission_handler/permission_handler.dart';
```

6. Android Configuration

AndroidManifest.xml

Path :

`android/app/src/main/AndroidManifest.xml`

Add permissions inside <manifest>:

```
<uses-permission android:name="android.permission.CAMERA"/>

<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>

<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>

<uses-permission android:name="android.permission.RECORD_AUDIO"/>
```

7. iOS Configuration

Info.plist

Path:

`ios/Runner/Info.plist`

Add:

```
<key>NSCameraUsageDescription</key>
<string>This app needs camera access.</string>

<key>NSLocationWhenInUseUsageDescription</key>
<string>This app needs location access.</string>

<key>NSMicrophoneUsageDescription</key>
<string>This app needs microphone access.</string>

<key>NSPhotoLibraryUsageDescription</key>
<string>This app needs photo library access.</string>
```

8. Check Permission Status

```
PermissionStatus status = await Permission.camera.status;

if(status.isGranted){
  print("Permission Granted");
}
```

- Permission.camera → Camera permission
- .status → Checks current status
- .isGranted → Returns true if allowed

9. Request Single Permission

```
PermissionStatus status =
    await Permission.camera.request();

if(status.isGranted){
    print("Camera Permission Granted");
}
```

- .request() shows permission popup
- User can Allow or Deny

10. Request Multiple Permission

```
Map<Permission, PermissionStatus> statuses =
    await [
        Permission.camera,
        Permission.location,
        Permission.storage,
    ].request();
```

11. Handle Permission Results

Granted :

```
if(status.isGranted){
    print("Access Granted");
}
```

Denied :

```
if(status.isDenied){
    print("Permission Denied");
}
```

12. Permanently Denied

```
if(status.isPermanentlyDenied){
    openAppSettings();
}
```

13. Camera Permission Example

```
Future<void> requestCamera() async {
  var status = await Permission.camera.request();

  if(status.isGranted){
    print("Camera Access Granted");
  } else {
    print("Camera Access Denied");
  }
}
```

14. Location Permission Example

```
Future<void> requestlocation() async {
  var status = await Permission.location.request();

  if(status.isGranted){
    print("Location Granted");
  }
}
```

15. Storage Permission Example

```
Future<void> requestStorage() async {
  var status = await Permission.storage.request();

  if(status.isGranted){
    print("Storage Granted");
  }
}
```

16. Permission Rationale Dialog

```
showDialog(
  context: context,
  builder: (context) {
    return AlertDialog(
      title: Text("Permission Required"),
      content: Text(
        "Camera permission is needed to take photos."
      ),
      actions: [
        TextButton(
          onPressed: (){
            Navigator.pop(context);
          },
          child: Text("OK"),
        ),
      ],
    );
  },
);
```

Code :

```
import 'package:flutter/material.dart';
import 'package:permission_handler/permission_handler.dart';

Run | Debug | Profile
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: PermissionScreen(),
    ); // MaterialApp
  }
}

class PermissionScreen extends StatelessWidget {
  const PermissionScreen({super.key});

  Future<void> requestPermission(
    BuildContext context, Permission permission, String name) async {

    PermissionStatus status = await permission.request();

    String message;
```

```
    if (status.isGranted) {
      message = "$name Permission Granted";
    } else if (status.isDenied) {
      message = "$name Permission Denied";
    } else if (status.isPermanentlyDenied) {
      message = "$name Permanently Denied";
      openAppSettings();
    } else {
      message = "$name Permission Status: $status";
    }

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text(message)),
    );
  }

  Widget permissionButton(
    BuildContext context, String title, Permission permission) {
    return Padding(
      padding: const EdgeInsets.all(8.0),
      child: ElevatedButton(
        onPressed: () {
          requestPermission(context, permission, title);
        },
        child: Text(title),
      ), // ElevatedButton
    ); // Padding
  }
}
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("All Permissions Demo"),
    ), // AppBar
    body: Center(
      child: SingleChildScrollView(
        child: Column(
          children: [
            permissionButton(
              context, "Camera", Permission.camera),
            permissionButton(
              context, "Location", Permission.location),
            permissionButton(
              context, "Microphone", Permission.microphone),
            permissionButton(
              context, "Contacts", Permission.contacts),
            permissionButton(
              context, "Storage", Permission.storage),
          ],
        ), // Column
      ), // SingleChildScrollView
    ), // Center
  ); // Scaffold
}
```

